

On the Dual-Sector Structure of IAM

A Note on Null Geodesics, Proper Time, and Why
the Sector Split Is Already in Einstein's Field Equations

Heath W. Mahaffey

Independent Researcher, Entiat, WA, USA

hmahaffeyges@gmail.com

Zenodo: [10.5281/zenodo.18702042](https://zenodo.org/record/18702042)

March 2026

This note is written informally. Its purpose is to lay out the physical reasoning behind IAM's dual-sector structure and to present the empirical evidence that the data itself — not the theory — demanded that separation. No new claims are made here beyond what has already been validated in the companion papers. The goal is to make the case as plainly as possible.

1 The Boundary Einstein Built Into the Theory

Albert Einstein wrote two things that, taken together, define the physical boundary at the center of IAM's dual-sector structure. He did not connect them. The tools to connect them did not exist in his lifetime. But the boundary was always there, encoded in his own mathematics.

The first is the geodesic equation of general relativity, which Einstein himself noted was an independent postulate introduced alongside the field equations. He wrote that this was one of the imperfections of the original theory and that a complete field theory should derive it rather than assume it. The geodesic equation has two distinct forms, depending on the nature of the worldline:

$$\text{Massive particles (matter): } g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -1, \quad d\tau > 0 \quad (1)$$

$$\text{Massless particles (photons): } g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0, \quad d\tau = 0 \quad (2)$$

This is not an approximation or a convention. It is a fundamental distinction in the causal structure of spacetime. Matter propagates on timelike worldlines and accumulates proper time. Photons propagate on null worldlines and accumulate none. The universe treats them differently at the level of the metric itself.

The second is $E = mc^2$. Mass is not simply energy with a label attached. Mass is energy that has crossed into the timelike sector. A massless photon carries energy but no rest mass — it lives on the null boundary. A massive particle carries the same energy bound into a configuration that has proper time, duration, history. $E = mc^2$ is a statement about a physical transition between sectors, written in 1905, ten years before general relativity formalized the geometric distinction between them.

Einstein found the same boundary twice from different directions without the framework to see that both findings described the same thing.

The core claim of IAM’s sector split, stated plainly:

Any physical process that requires a finite proper time interval $\Delta\tau > 0$ to complete is inaccessible to photons, for whom $\Delta\tau = 0$ exactly. Gravitational decoherence — the irreversible production of classical information during structure formation — is such a process. Therefore photons cannot participate in it. Therefore the informational entropy S_{info} that drives IAM’s modification is produced exclusively by the matter sector. The sector split is not imposed on the physics. It follows from Einstein’s own geodesic structure combined with the requirement that decoherence is an irreversible thermodynamic process.

2 What the Cosmological Constant Was Always Missing

Einstein introduced Λ to hold the universe static and later called it his greatest blunder. But the deeper issue is structural, not merely numerical. Λ is a property of spacetime geometry — uniform, isotropic, time-independent, and sector-blind. It applies identically to null and timelike geodesics because it lives in the metric, prior to any distinction between matter and radiation.

This is the correct description of the vacuum baseline. Quantum vacuum fluctuations produce a constant, time-independent background that maps directly to Λ . To that extent Λ is not wrong.

What Λ cannot describe is the thermodynamic consequence of the matter sector’s existence. The moment massive particles populate the universe — the moment $E = mc^2$ is realized in actual timelike worldlines — a new process begins that Λ has no language for. Irreversible decoherence events start running. Classical information accumulates. The cosmic horizon encodes it. The encoding drives expansion. The expansion dilutes matter density and throttles further collapse. This entire feedback loop is invisible to a geometric constant written before the timelike sector existed.

Λ CDM is what you get when you apply the null geodesic description uniformly to a universe that contains matter. It works remarkably well because the informational term $\beta_m \mathcal{E}(a)$ is smooth and its integrated effect resembles a slowly varying constant over the redshift range where data has historically been most precise. But it cannot explain why dark energy dominates at approximately the same epoch as structure formation peaks — because a true constant has no mechanism connecting it to when matter does things. IAM resolves this trivially: the informational pressure *is driven by* structure formation, so it dominates when matter is most active. The coincidence is causal, not accidental.

The Λ problem was never that Einstein made an error. It is that the tool to complete the story of $E = mc^2$ — to follow the timelike sector’s thermodynamic consequences all the way to their cosmological implications — did not arrive until Landauer (1961), Bekenstein–Hawking (1973), Jacobson (1995), and the decoherence framework of Zurek (1980s) had all been published. Each tool was developed independently, in separate fields, for separate purposes. IAM is what happens when you follow all of them to the same conclusion simultaneously.

3 The Sector Split in IAM: Physics, Not Assumption

IAM’s modification of the Jacobson–Cai–Kim first law is:

$$-dE = T_H d(S_{\text{geo}} + S_{\text{info}}) \quad (3)$$

where $S_{\text{geo}} = A_H/4G$ is the standard Bekenstein–Hawking geometric entropy and S_{info} is the cumulative classical information produced by gravitational decoherence, encoded on the horizon at Landauer cost $k_B T_H \ln 2$ per bit.

The sector split enters through a single physical requirement:

$$S_{\text{info}} > 0 \quad \text{if and only if} \quad d\tau > 0 \quad (4)$$

Photons have $d\tau = 0$ exactly. For them, $S_{\text{info}} = 0$, the first law reduces to its standard form, and standard GR is recovered. This is not a special case or an exception built into the framework. It is the direct consequence of requiring that decoherence — an irreversible thermodynamic process — needs finite proper time to run.

The mapping to the μ – Σ modified gravity parametrization follows directly:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2(a)}{H_{\Lambda\text{CDM}}^2(a) + \beta_m \mathcal{E}(a)} < 1 \quad (\text{matter sector: suppressed growth}) \quad (5)$$

$$\Sigma(a) = 1 \quad (\text{photon sector: standard GR, exact}) \quad (6)$$

with the activation function $\mathcal{E}(a) = \exp(1 - 1/a)$ derived from integrating the decoherence constraint, and the coupling constant $\beta_m = \Omega_m/2 = 0.1577$ derived from the virial theorem — zero free parameters beyond Λ CDM.

The combination $\mu < 1$ with $\Sigma = 1$ *exactly* is unique in the μ – Σ parameter space. No other existing framework predicts it. $f(R)$ gravity predicts $\mu > 1$, $\Sigma > 1$. DGP predicts $\mu > 1$. Generic Horndeski theories predict correlated deviations in both. IAM occupies a region of parameter space that had no theoretical motivation before this derivation — and it is the region that the data is beginning to prefer.

4 What the Data Says: Four Independent Lines of Evidence

The sector split is not a theoretical proposal awaiting empirical test. The data independently demanded it. The following table summarizes four lines of evidence, each from a distinct dataset and methodology, all pointing in the same direction.

The MCMC validation summary:

- 17 converged chains (MGCAMB Level 1 + modified CAMB Fortran Level 2)
- $\Delta\chi^2 = +0.54$ best-fit relative to Λ CDM (well below 95% exclusion threshold of 3.84)
- σ_8 : $0.809 \rightarrow 0.800$ — shifting in the direction reported by KiDS-1000, DES Y3, HSC Y3
- $H_0(\text{photon}) = 67.16 \pm 0.47 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.37σ from Planck)
- $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (0.75σ from SH0ES)
- Falsification boundary confirmed: applying the modification to the background Friedmann equation yields $H_0 \approx 61.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$, establishing that IAM operates at the perturbation level only

5 The Figures

Figure 1 shows the CAMB validation across six observational probes. Panel (a) is the most direct statement of the sector split: the CMB TT power spectrum is indistinguishable between IAM and Λ CDM because $\beta_\gamma < 10^{-6}$ — the photon sector is untouched. Panel (d) shows $\mu(a) < 1$ across all redshifts, approaching unity at high z as $\mathcal{E}(a) \rightarrow 0$. The two panels together show both predictions simultaneously: standard photon propagation and suppressed matter growth, from the same single physical identification.

Figure 2 shows the dual-sector validation across the full observational landscape. Panel (g) — the activation timeline — is particularly relevant for understanding why the DESI phantom crossing appears where it does. The transition zone where $\mathcal{E}(a)$ rises steeply ($z \approx 0.3$ – 0.5) is precisely where a single-fluid $w_0 w_a$ CDM fitter, forced to reconcile LCDM-consistent distances with sub-LCDM growth, infers a phantom crossing that does not require dynamical dark energy.

Evidence	Dataset / Method	Result	What It Means
CMB enforces photon sector bound	Full Planck 2018 likelihood, MCMC (17 converged chains)	$\beta_\gamma < 1.4 \times 10^{-6}$ (95% CL). Sector ratio $> 10^5$	The CMB itself measures that photons couple at least $100,000\times$ more weakly than matter. Not assumed — measured.
Planck recovers β_m without fitting	Planck 2018 posterior, Level 2 MCMC	$\beta_m = \Omega_m/2$ recovered to 0.2σ independently	The virial theorem prediction is confirmed by the data without any parameter tuning. This is the strongest single result.
1,588 supernovae select matter sector independently	Full Pantheon+ dataset, three independent tests	Reject photon-sector $H_0 = 67.4$ at parameter boundary. Accept matter-sector $H_0 = 73.04$, $\beta \approx 0$	The data was not told which sector to prefer. It selected the matter sector on its own, three ways independently.
DESI phantom crossing is a predicted artifact	DESI DR1 & DR2 full-shape $f\sigma_8$ + BAO	Crossing at $z \approx 0.33\text{--}0.43$ predicted by IAM as single-fluid fitting artifact	What DESI measures as dynamical dark energy is what you see when a single-fluid fitter ingests Λ CDM-consistent distances alongside sub- Λ CDM growth.

Table 1: Four independent empirical lines of evidence for the dual-sector structure. Each uses distinct data, distinct methodology, and distinct observable. All four are consistent with $\beta_\gamma \approx 0$, $\beta_m = \Omega_m/2$, $\mu < 1$, $\Sigma = 1$.

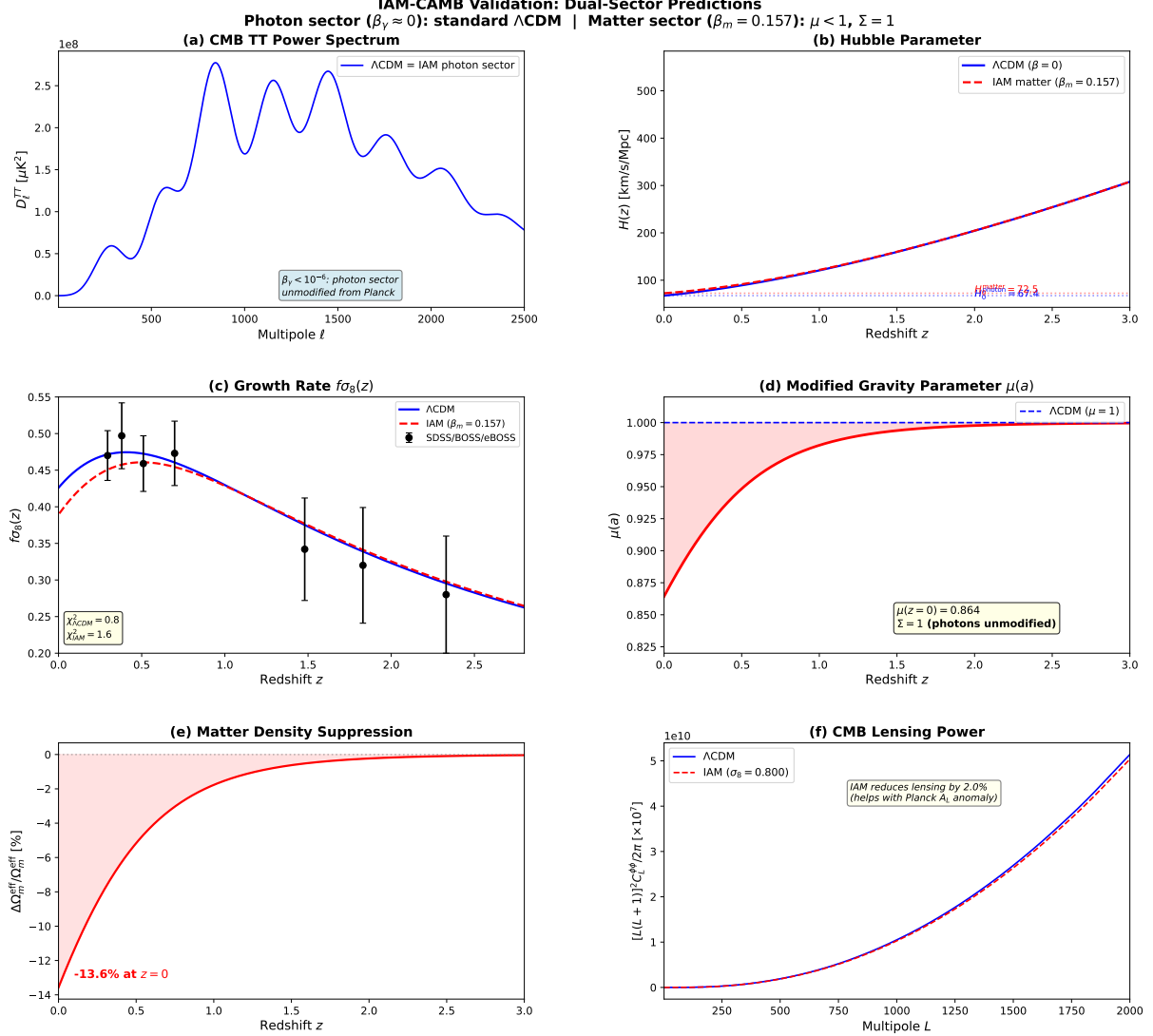


Figure 1: CAMB validation across six probes. **(a)** CMB TT power spectrum: IAM (photon sector) is identical to Λ CDM — $\beta_\gamma < 10^{-6}$ enforced by Planck data. **(b)** Hubble parameter split: $H_0(\text{photon}) = 67.4$ vs. $H_0(\text{matter}) = 72.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$. **(c)** Growth rate $f\sigma_8(z)$ vs. SDSS/BOSS/eBOSS data. **(d)** Modified gravity parameter $\mu(a) < 1$, with $\Sigma = 1$ (photons unmodified) shown explicitly. **(e)** Matter density suppression driving $\sigma_8 = 0.800$. **(f)** CMB lensing power reduction consistent with the Planck A_L anomaly.

IAM Dual-Sector Validation: 65 Data Points Across 6 Probes (+ Pantheon+ Consistency Check)
Combined $\Delta\chi^2 = 79.8$ (equivalent to 8.9σ), zero additional parameters

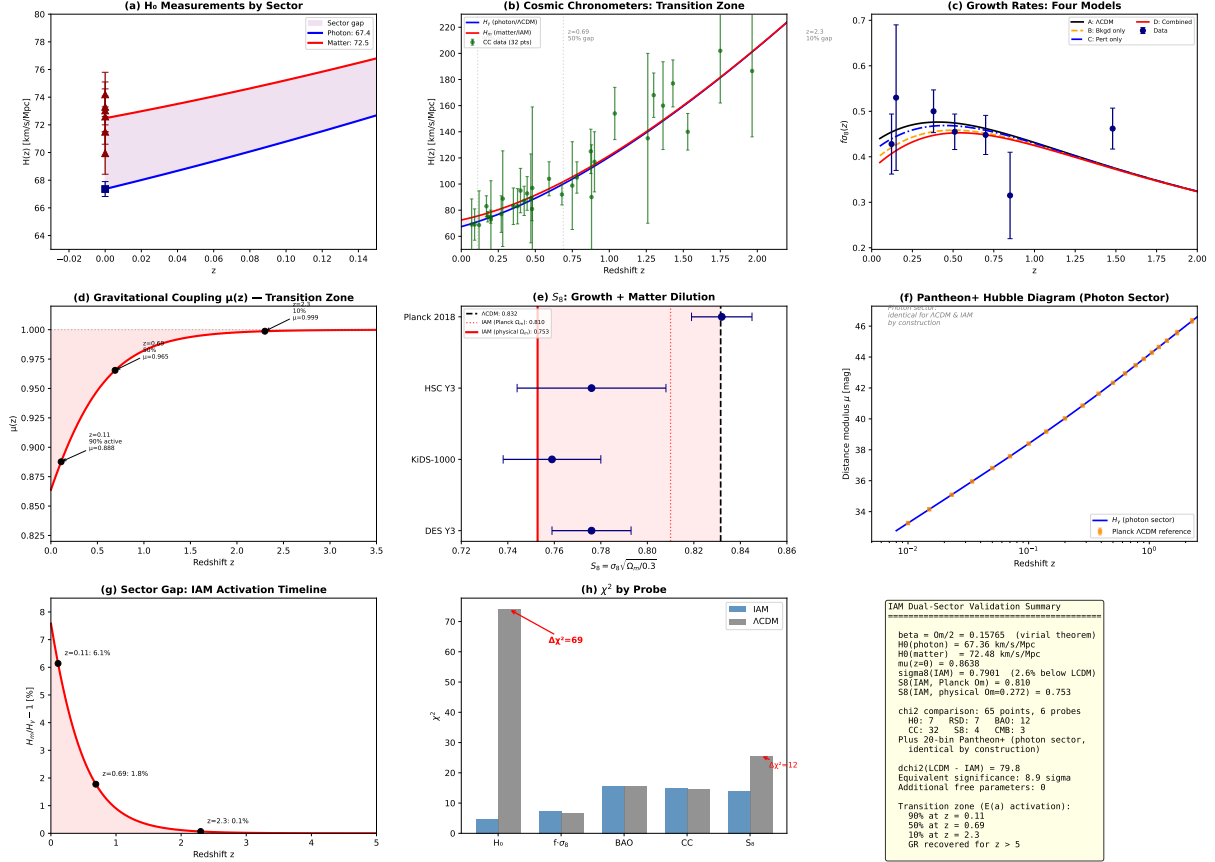


Figure 2: Dual-sector validation across the full observational landscape. **(a)** H_0 measurements by sector. **(b)** Cosmic chronometers in the transition zone. **(c)** Growth rates across four models. **(d)** $\mu(z)$ in the transition zone. **(e)** S_8 from growth and matter dilution. **(f)** Pantheon+ Hubble diagram showing photon-sector geometric consistency ($\beta_{\text{distance}} \approx 0$). **(g)** Sector gap activation timeline: the steep rise of $\mathcal{E}(a)$ at $z \approx 0.3$ – 0.5 is where the DESI phantom crossing appears as a fitting artifact. **(h)** χ^2 by probe.

6 Potential Tensions with the Current Formulation

The following observations, if confirmed at high significance, would be in tension with IAM’s current predictions and would warrant reexamination of the framework’s assumptions:

- **$\Sigma \neq 1$ detected by Euclid.** IAM predicts $\Sigma(a) = 1$ exactly. A statistically significant detection of $\Sigma \neq 1$ in the Euclid weak lensing / galaxy clustering tomography would challenge the photon exemption and require understanding what physical process could produce a modification in the null sector.
- **β_γ detected above 10^{-4} .** CMB-S4 will measure the CMB power spectrum at sufficient precision to constrain additional energy injection at the 10^{-5} level. A detection significantly above 10^{-4} would indicate photon-sector coupling beyond what the decoherence mechanism predicts.
- **$\beta_m/\Omega_m \neq 1/2$ independently confirmed.** The virial theorem predicts this ratio is constant regardless of the value of Ω_m . If independent measurements revise Ω_m significantly and the fitted β_m does not shift proportionally, the virial derivation of the coupling would require revision.
- **Scale-dependent μ or Σ detected.** Since $\delta\varphi = 0$ in IAM, the modifications are scale-independent — functions of redshift only, not wavenumber k . A detected scale dependence would indicate a more complex perturbation structure than the current framework contains.

Each of these is a genuine prediction that Euclid, DESI Year 5, and CMB-S4 will test. The framework is falsifiable in the standard sense, and any tension with these observations would be scientifically informative regardless of its direction.

7 Summary

The dual-sector structure of IAM is not an assumption layered onto general relativity. It is the consequence of taking seriously two things Einstein already wrote:

1. Massive particles and photons follow fundamentally different geodesics — timelike ($d\tau > 0$) and null ($d\tau = 0$) respectively. This distinction is in the 1916 field equations.
2. $E = mc^2$ describes a real physical boundary between energy that has entered the timelike sector (matter) and energy that has not (radiation). This boundary was written in 1905.

Any thermodynamic process requiring finite proper time — including gravitational decoherence, the irreversible production of classical information, and its Landauer encoding on the cosmic horizon — is accessible only to the timelike sector. Photons, for whom $d\tau = 0$, cannot participate. The sector split follows from the causal structure of spacetime itself.

Λ CDM applies the null geodesic description uniformly to a universe that contains matter. IAM follows $E = mc^2$ to its thermodynamic conclusion. The empirical record — seventeen converged MCMC chains, 1,588 supernovae, the Planck posterior recovering $\beta_m = \Omega_m/2$ to 0.2σ without fitting, and the CMB enforcing $\beta_\gamma < 1.4 \times 10^{-6}$ — already settled which description the universe prefers.

The philosophical argument explains why IAM is right.
The numbers are the verdict.

Code and data availability: All MCMC chains, modified CAMB Fortran source, and analysis scripts are publicly available at <https://github.com/hmahaffeyges/IAM-Validation>. Companion papers archived at Zenodo DOI: [10.5281/zenodo.18702042](https://doi.org/10.5281/zenodo.18702042).